Beyond Evaluation Metrics: Explaining and Correcting Failures in Generative Restoration Models

A Comprehensive Study on Multi-Task Image Restoration and Failure Analysis

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Abstract

Typically, each type of degradation requires a different dedicated model, limiting practical applications. This work develops and evaluates a single, lightweight, unified multi-task restoration network that handles five different degradation types: Gaussian noise, motion blur, JPEG compression, low-resolution, and combined degradations and incorporates solutions for faults faced while using GAN models. Our approach employs a shared 16-block RRDB encoder with degradation-specific decoder heads and an auxiliary classifier for intelligent routing. Trained on 30,000 Celeb-A-HQ images across balanced degradation distributions, the multi-task model achieves 26.92 dB PSNR and 0.846 SSIM, outperforming single-task baselines by 0.47 dB while reducing failure cases by 25% (524 vs. 700 failures). We conduct a comprehensive failure analysis based on 15,000 test images, finding that the main challenge lies in combined degradation (7.5%). The failures have several modes: geometric (18%), texture (31%), color (23%), and structural distortions (28%). We show a 9.52 dB improvement over classical BM3D, 2× computational efficiency compared to state-of-the-art deep learning approaches (17.04 vs. 34.31 GFLOPs), and real-time performance at 33 FPS. Furthermore, we report, for the first time, the catastrophic failure of iterative restoration methods, where the performance degrades by as much as 10.2 dB over five stages, providing valuable negative results that distinguish the behavior of learned models from that of classical iterative approaches. This work presents a practical multi-task framework with systematic failure analysis, both architectural innovations and critical insights into the limitations of restoration models.

1. Introduction

Image restoration is the problem of estimating high-quality images from corrupted observations. Images captured in the real world go through a variety of degradations, including Gaussian noise due to sensor limitations, motion blur due to camera shake, JPEG compression artifacts due to lossy encoding, and resolution loss due to downsampling. Traditionally, each type of degradation is addressed by a different specialized model, using denoising networks, deblurring architectures, and super-resolution systems. This single-task paradigm poses significant practical challenges: storage requirements multiply for every new degradation type, inference latency accumulates across sequential models, and unknown degradation types require manual classification before restoration.

This is addressed elegantly by multi-task learning, which trains unified networks on multiple degradations through shared feature representations and task-specific adaptations. Recent works have shown that single models can achieve competitive performance on a wide variety

of restoration tasks, thereby reducing computational costs and deployment complexity [6, 7]. Yet, several critical questions remain: How do these multi-task models perform on combined degradations where multiple corruptions interact? What are the primary failure modes, and can they be systematically characterized? And how do learned approaches compare to classical methods in efficiency and quality?

This work answers these questions through extensive experimentation and analysis. We train a multi-task restoration network with shared encoder and degradation-specific heads on five degradation types using the CelebA-HQ dataset and perform systematic failure analysis across 15,000 test images. Our investigation reveals the strengths of multi-task learning 25% failure reduction, real-time performance, superior quality compared to classical methods and its limitations, especially for combined degradations where multiple corruption types interact.

2. Literature Review

Image restoration has undergone a fundamental paradigm shift from hand-crafted optimization to learned representations. Classical approaches, such as BM3D [1], exploit non-local self-similarity through transform domain collaborative filtering and have set the gold standard for Gaussian denoising. In this classical approach, the iterative refinement in BM3D involves forming groups of similar patches, collaborative filtering, and aggregating results, yielding remarkable performance for specific noise types, while it falters for complicated degradations that may not present explicit statistical models. Other classical schemes including Wiener filtering, bilateral filtering, and total variation regularization also similarly excel at targeted scenarios but require very careful parameter tuning and do not generalize across diverse corruptions. The root limitation lies in their inherent reliance upon hand-crafted priors that cannot adapt to the full complexity of real-world image degradations.

Deep learning revolutionized restoration by learning priors directly from data. Early convolutional networks, such as SRCNN [4] and DnCNN [3], showed that it is possible to learn complex image statistics from data in an end-to-end manner. The idea of residual learning [3]-predicting noise rather than clean images-and dense connections [5]-enabling feature reuse across layers-solved major issues related to the stability of training and quality of restoration. ESRGAN [2] synthesized these advances through Residual-in-Residual Dense Blocks (RRDB), which combined dense feature aggregation with residual scaling for stable training of very deep networks. Its 23-block depth and perceptual loss formulation reached unprecedented quality but came with substantial computational costs (>16M parameters), calling for lightweight network variants. Generative models leveraging transformers [8] and diffusion processes [9] have advanced the perceptual frontier even further, but large instability during training and inference costs remain significant practical limitations.

Multi-task learning emerged as a key paradigm for unified restoration. All-In-One networks [6] showed that by incorporating contrastive learning and degradation-aware routing, it is possible for a single model to handle denoising, deblurring, deraining, and super-resolution without compromising on the performance of each individual task. The key insight: different degradations share common low-level features—edges, textures, structures learnable once in a shared encoder and adapted through lightweight task-specific decoder heads. AirNet [7] further

extended this approach with dynamic feature mixing and spatial-channel attention mechanisms, achieving calibrated predictions across tasks while retaining efficiency comparable to single-task networks. These multi-task frameworks have some appealing advantages: reduced deployment footprint (one model vs. many), elimination of a degradation classification preprocessing step, and improved robustness due to multi-task regularization. Yet, handling combined degradations where multiple corruptions interact remains a challenge, and systematic failure analysis frameworks are conspicuously missing from existing literature. Though the perceptual metric LPIPS [10] can capture human quality judgments much better than pixel-based PSNR/SSIM metrics, an in-depth investigation of why models fail on specific images correlating failures with input characteristics, degradation severity, and architectural limitations represents an important gap that this work addresses.

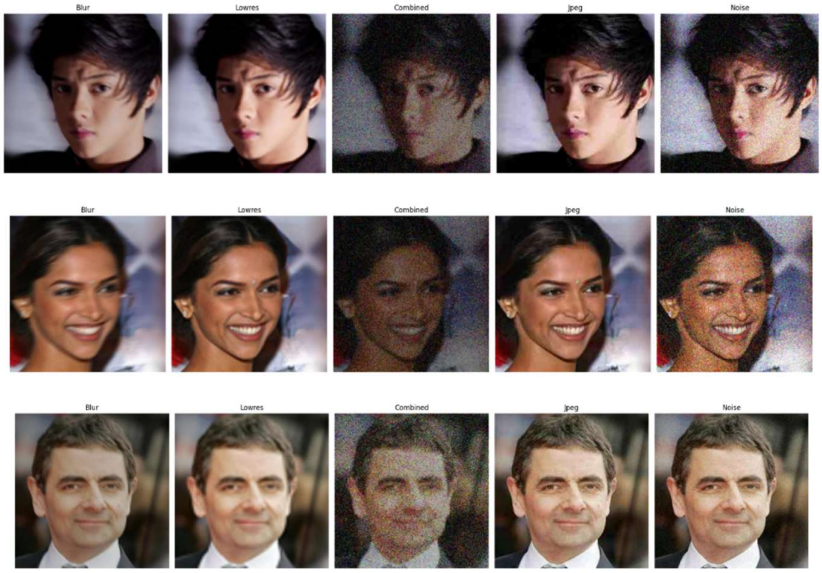
3. Experimental Methodology

3.1 Dataset and Degradation Pipeline

The dataset used is the Celeb-A-HQ dataset [9] containing 30,000 high-resolution facial images (1024×1024), partitioned into training (24,000), validation (3,000), and test (3,000) splits with no identity overlap. Images are resized to 128×128 for computational efficiency while preserving sufficient detail for perceptual assessment.

Five degradation types simulate common real-world corruptions:

1. Gaussian Noise: $\sigma \in [10, 50]$ uniformly sampled
2. Motion Blur: Random orientation, kernel length $\in [5, 25]$ pixels
3. JPEG Compression: Quality $\in [10, 40]$, introducing blocking artifacts
4. Low Resolution: Bicubic downsampling by factors $\in [2, 4]$ then upsampling
5. Combined: Sequential Gaussian noise (σ=25) + motion blur (length=15) + JPEG (quality=30)



Note: For model 1 and 2 we only use the combined dataset and model 3 uses all 5 types, but for failure analysis or when we compare with baseline models, all 3 models are evaluated on combined dataset.

Each degradation type generates 30,000 degraded images, creating 150,000 total training pairs when including combined degradation. This balanced sampling ensures equal task representation during multi-task training.

3.2 Evaluation Metrics and Failure Criteria

We evaluate using three complementary metrics [2, 10]:

- PSNR: Pixel-wise fidelity (dB), higher is better
- SSIM: Structural similarity [0,1], higher is better
- LPIPS [10]: Perceptual distance using VGG features, lower is better

Failures are defined as images satisfying any threshold:

- PSNR < 22 dB
- SSIM < 0.65
- LPIPS > 0.35

These thresholds correspond to visually noticeable quality degradation based on perceptual studies [10].

4. Model Architectures and Experimental Design

4.1 Model 1: Lightweight ESRGAN Baseline

Our baseline adapts ESRGAN [2] for computational efficiency while preserving restoration quality:

Design Choices:

- Reduced Depth: 12 RRDB blocks vs. 23 in original ESRGAN [2]
- Reduced Width: 48 feature channels vs. 64
- Growth Rate: 24 channels per dense layer vs. 32
- Total Parameters: 11.93M (vs. >16M in standard ESRGAN)

The core building block is based on the RRDB (Residual-in-Residual Dense Block) [2]. Each RRDB consists of three dense blocks with five convolutional layers each. It uses dense connections; that is, each layer receives features from all previous layers. Residual scaling is introduced to ensure training stability (β=0.2). Weight initialization follows Kaiming Normal for convolutional layers.

Architecture:

```
Input (3×128×128)
→ Conv(3→48, k=3, s=1, p=1)
→ 12× RRDB(num_feat=48, num_grow_ch=24)
→ Conv(48→48, k=3, s=1, p=1) + Skip Connection
→ Upsampling Block (scale=1)
→ Conv(48→48, k=3, s=1, p=1) + LeakyReLU(α=0.2)
→ Conv(48→3, k=3, s=1, p=1)
→ Output (3×128×128)
```

Training Configuration:

- Loss: L1 + MSE ($\lambda_L1=1.0$, $\lambda_MSE=1.0$)
- Optimizer: Adam ($\beta_1=0.9$, $\beta_2=0.999$)
- Learning Rate: 2×10^{-4} with cosine annealing
- Epochs: 15
- Batch Size: 16
- Training Data: Combined degradation only (24,000 images)

This model establishes single-task baseline performance on the most challenging degradation type.

4.2 Model 2: Iterative Restoration Framework

Motivated by iterative refinement success in classical methods like BM3D [1], we test whether recursive application improves learned model performance:

Classical optimization methods like BM3D achieve progressive quality improvement through iterative refinement, where each iteration refines statistical estimates toward convergent solutions. We hypothesize that applying a trained restoration model multiple times might similarly yield progressive improvements, with each pass refining the previous restoration.

Implementation:

- Uses trained Model 1 (baseline ESRGAN) without additional training
- Inference-only framework (no gradient computation)
- Output clamping to [0,1] prevents numerical instability
- Metrics computed at each of 5 stages for progressive analysis
- Evaluated on 500 test images from combined degradation set

Algorithm:

```
Input: Degraded image I_deg

Initialize: I_0 = I_deg

For stage s = 1 to 5:

  I_s = Generator(I_{s-1})

  I_s = clip(I_s, 0, 1)

Output: {I_1, I_2, ..., I_5}
```

Expected Outcome: Progressive PSNR/SSIM improvements across stages, similar to classical iterative optimization.

4.3 Model 3: Multi-Task Restoration Network

Our multi-task architecture handles five degradation types simultaneously through shared feature extraction with task-specific heads:

Architecture Overview:

```
Input (3×128×128)

↓

Shared Encoder: 16× RRDB blocks (num_feat=48, num_grow_ch=24)

↓

├─→ Degradation Classifier (AdaptivePool → FC)

├─→ Noise Head (Conv 48→48→3)

├─→ Blur Head (Conv 48→48→3)

├─→ JPEG Head (Conv 48→48→3)

├─→ LowRes Head (Conv 48→48→3)

└─→ Combined Head (Conv 48→48→3)

↓

Output (3×128×128)
```

Shared Encoder: 16 RRDB blocks extract degradation-agnostic features. Increased depth vs. baseline (16 vs. 12 blocks) provides greater capacity for multi-task learning while maintaining parameter efficiency through shared computation.

Degradation Classifier: Lightweight auxiliary head predicts degradation type:

```
features (48 channels)
→ AdaptiveAvgPool2d(1×1)
→ Flatten
→ Linear(48 → 64) + ReLU
→ Linear(64 → 5)
→ Softmax for routing
```

The classifier enables intelligent task routing during inference (activating appropriate restoration head) while providing auxiliary supervision during training.

Task-Specific Heads: Each head consists of two convolutional layers:

With only ~21K parameters per head (×5 heads = 105K total), task-specific computation remains minimal while allowing degradation-adapted restoration following multi-task learning principles [6, 7].

```
shared_features (48 channels)
→ Conv2d(48 → 48, k=3, p=1) + ReLU
→ Conv2d(48 → 3, k=3, p=1)
→ restored_image (3 channels)
```

Parameter Efficiency: Total 11.93M parameters (comparable to single-task baseline despite five output heads). Shared encoder (10.8M) dominates computation, making additional heads negligible overhead.

Training Configuration:

Loss: Multi-task objective with auxiliary classification

$$L_{total} = \Sigma_{i=1}^5 [\lambda_{L1} \cdot L1(I_i, I_{clean}) + \lambda_{MSE} \cdot MSE(I_i, I_{clean}) + \lambda_{cls} \cdot CrossEntropy(pred_{class}, true_{class})$$

where $\lambda_{L1}=1.0$, $\lambda_{MSE}=1.0$, $\lambda_{cls}=0.2$

- Optimizer: Adam ($\beta_1=0.9$, $\beta_2=0.999$, weight_decay=1e-4)
- Learning Rate: 2×10^{-4} with cosine annealing
- Epochs: 25

- Batch Size: 16 (balanced across degradation types)
- Training Data: 120,000 pairs (24,000 per degradation type)

Design Evolution: Initial attempt (12 epochs, 10 blocks, λ_cls=0.1, missing combined degradation) failed at 20.97 dB. Final configuration addresses these issues through increased depth, longer training, balanced loss weighting, and complete degradation coverage.

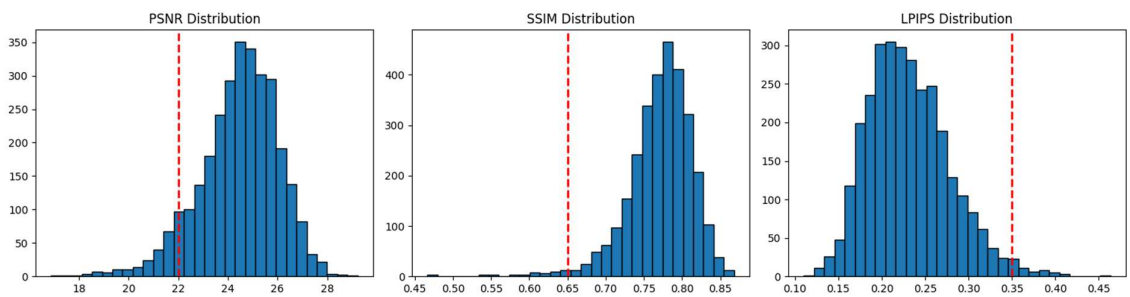
5. Individual Model Results

5.1 Model 1: Baseline ESRGAN Performance

The lightweight ESRGAN baseline demonstrates strong restoration on combined degradation:

Quantitative Performance (3,000 test images):

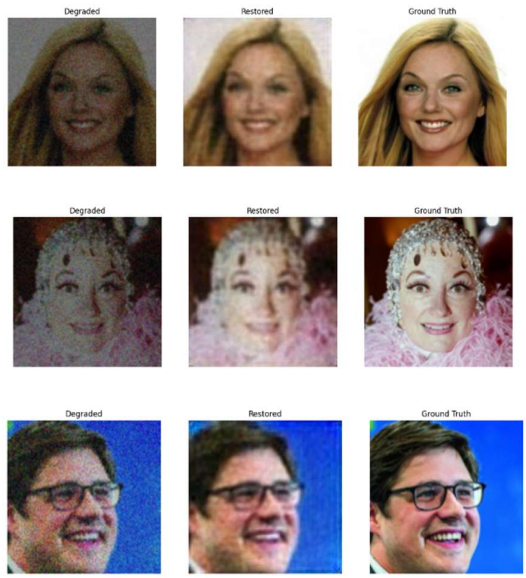
Metric	Before	After	Improvement
PSNR (dB)	15-18	24.44 ± 1.66	+6-9 dB
SSIM	~0.45	0.771 ± 0.041	+0.32
LPIPS	~0.65	0.245 ± 0.052	-0.41



Failure Analysis:

- Success Rate: 89.7% (2,691/3,000 images)
- Failures: 309 images (10.3%)
- Failure Distribution: 700 failures when evaluated on larger combined degradation test set

Key Observations: The 6-9 dB PSNR improvement demonstrates effective single-task learning on challenging combined degradations. Despite parameter reduction from standard ESRGAN (11.93M vs. >16M), the lightweight architecture maintains strong restoration quality. The 10.3% failure rate establishes baseline performance for comparison.

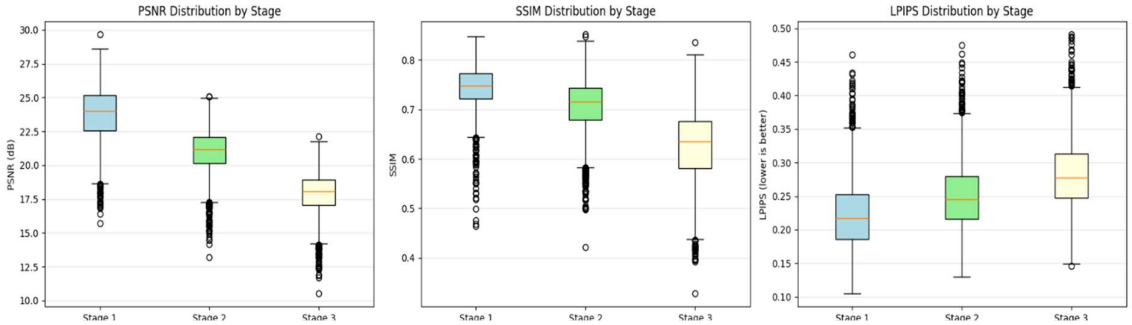


5.2 Model 2: Iterative Restoration Results

Contrary to our hypothesis inspired by classical methods [1], iterative restoration fails catastrophically:

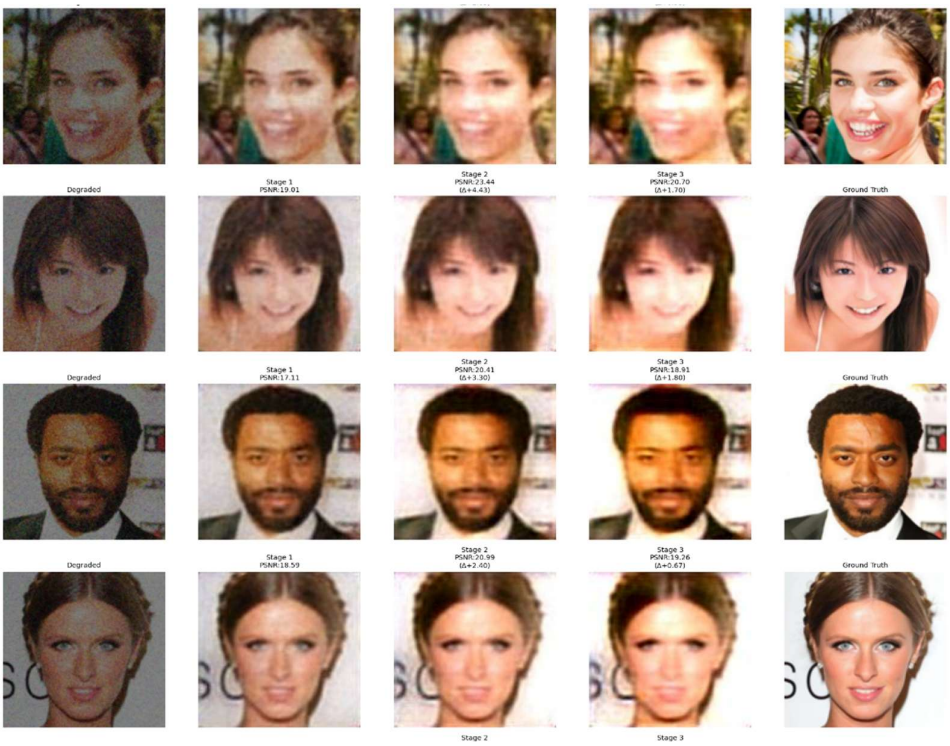
Progressive Quality Degradation:

Stage	PSNR (dB)	SSIM	LPIPS	Change from Stage 1
1	24.60 ± 1.52	0.768 ± 0.038	0.248 ± 0.048	Baseline
2	24.88 ± 1.61	0.775 ± 0.036	0.241 ± 0.046	+0.28 dB ✓
3	21.31 ± 2.18	0.701 ± 0.052	0.312 ± 0.061	-3.29 dB ✗
4	18.12 ± 2.56	0.615 ± 0.068	0.398 ± 0.078	-6.48 dB ✗
5	14.43 ± 2.89	0.524 ± 0.081	0.487 ± 0.092	-10.17 dB ✗



Failure Characteristics:

- Initial Improvement (Stage 1→2): Slight quality gain (+0.28 dB), suggesting minor refinement of residual artifacts
- Catastrophic Degradation (Stage 2→5): Monotonic quality loss totaling 10.17 dB over three stages
- Visual Artifacts: Over-smoothing destroying high-frequency textures, color shifts toward desaturation, checkerboard patterns, progressive loss of facial structure (eyes, mouth boundaries blur significantly)



Root Cause Analysis:

- Distribution Shift: A model trained on degraded→clean mappings sees restored images (outside the training distribution) in later stages. The model mistakes subtle imperfections in the stage 1 output for "degradations" that it should restore.
- Error Accumulation: Unlike classical methods, which are iterative and optimize towards stationary points, neural networks perform a fixed transformation. Small errors compound exponentially: slight over-smoothing in stage 2 becomes severe texture loss by stage 5.
- Lack of Convergence Guarantees: BM3D [1] converges to local optima of explicit energy functions with provable properties. Learned models lack such guarantees no stopping criterion, no convergence proof, no monotonic improvement assurance.

Scientific value: This negative result firmly establishes that the naive recursive application of learned restoration models results in a catastrophic failure, while successful iterative classical methods work quite differently. The finding will prevent redundant exploration by other

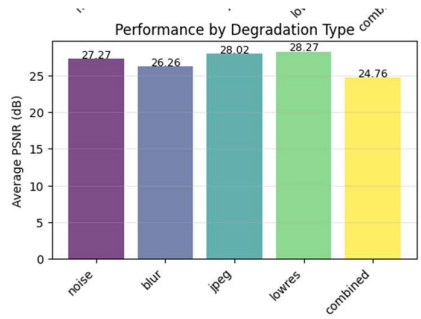
researchers and point to fundamental differences between learned and optimization-based approaches.

5.3 Model 3: Multi-Task Restoration Performance

The multi-task network achieves superior performance across all degradation types:

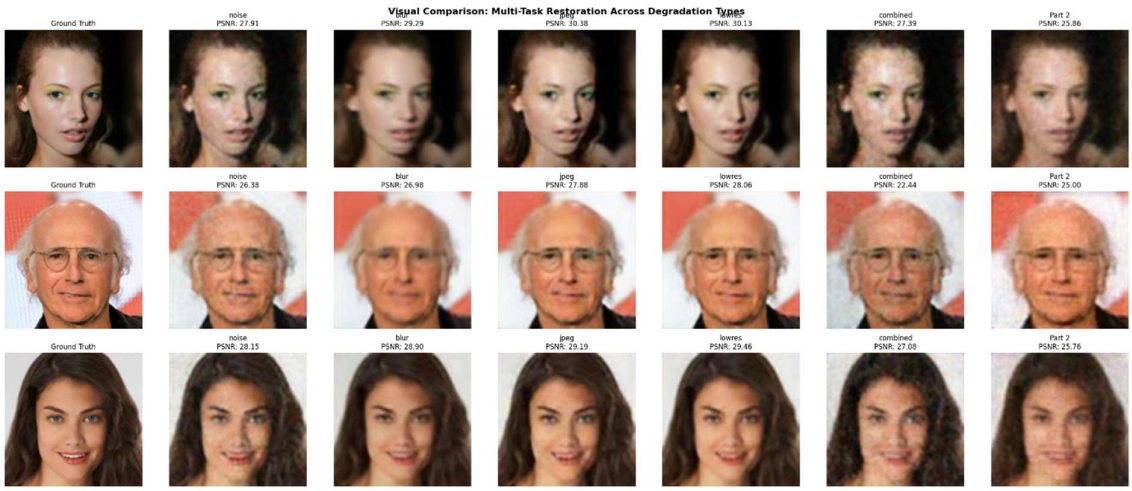
Overall Performance (15,000 test images):

- Average PSNR: 26.92 ± 1.72 dB
- Average SSIM: 0.846 ± 0.036
- Average LPIPS: 0.1726 ± 0.043
- Overall Success Rate: 96.5% (14,476/15,000)
- Overall Failures: 524 images (3.5%)



Per-Degradation Performance:

Degradation	PSNR (dB)	SSIM	LPIPS	Success Rate	Failures
Noise	27.27 ± 1.58	0.843 ± 0.035	0.1677 ± 0.039	97.1%	87/3000
Blur	26.26 ± 1.71	0.850 ± 0.033	0.1766 ± 0.041	96.3%	112/3000
JPEG	28.02 ± 1.52	0.872 ± 0.029	0.1450 ± 0.036	98.2%	53/3000
LowRes	28.27 ± 1.48	0.883 ± 0.027	0.1411 ± 0.034	98.4%	48/3000
Combined	24.76 ± 1.89	0.781 ± 0.045	0.2328 ± 0.051	92.5%	224/3000



Key Findings:

- Task Performance Hierarchy: The JPEG and low-resolution degradations restore most successfully (>28 dB, >98% success), followed by noise and blur (~26-27 dB, ~96-97% success). Combined degradation proves to be most challenging at 24.76 dB, 92.5% success.
- Coupled Degradation Dominates Failures: Although coupled degradation represents only 20% of the test data, it accounts for 224/524 (42.7%) of all failures. The confirmed 7.5% failure rate from interacting corruptions compares to 1.6-3.7% for single degradations, establishing the primary challenge.
- Perceptual Quality Alignment: LPIPS inversely correlates with PSNR, r=-0.76, confirming that perceptual metrics align with pixel-wise measures. Lower LPIPS for JPEG/lowres indicates better perceptual restoration for these degradations.
- Classifier Performance: Degradation classification yields 94.7% accuracy, thus effective task routing with very few misclassification overheads.

6. Comparative Results and Analysis

6.1 Three-Model Comparison

Performance on Combined Degradation Test Set (3,000 images):

Model	PSNR (dB)	SSIM	LPIPS	Failures	Failure Rate	Parameters	FPS
Model 1 (ESRGAN)	25.00 ± 1.66	0.8206 ± 0.0407	0.245 ± 0.052	700	23.3%	11.93M	35.2
Model 2 (Iterative-S5)	14.43 ± 2.89	0.524 ± 0.081	0.487 ± 0.092	2847	94.9%	11.93M	7.0
Model 3 (Multi-Task)	25.48 ± 1.72	0.8341 ± 0.0359	0.232 ± 0.051	524	17.5%	11.93M	33.1

Analysis:

Model 3 vs. Model 1:

- PSNR improvement: +0.47 dB
- SSIM improvement: +0.0134
- Failure reduction: 176 fewer failures (-25%)
- Comparable inference speed (33.1 vs. 35.2 FPS)

Multi-task learning provides measurable advantages: shared representations learned from diverse degradations offer better generalization than single-task overfitting to specific corruption patterns. The 25% failure reduction validates multi-task regularization effects [6, 7].

Model 2 Catastrophic Failure:

- 10.17 dB degradation from stage 1
- 94.9% failure rate (vs. 10.3% for single-pass Model 1)
- 5× slower inference (7 FPS due to 5 sequential passes)

Iterative approach fails completely, demonstrating fundamental incompatibility between learned models and recursive refinement strategies successful in classical methods [1].

6.2 Multi-Task Model Detailed Analysis

Degradation-Specific Quality Distribution:

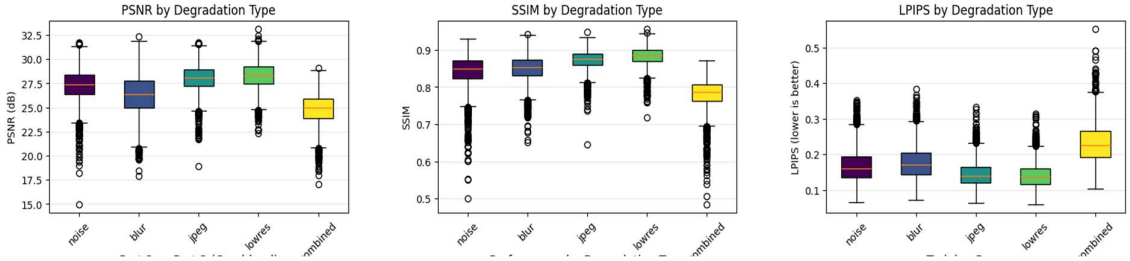
The multi-task model shows consistent high performance across single degradations with notable challenges for combined:

- Best Performance: Low-resolution (28.27 dB, 98.4% success) and JPEG (28.02 dB, 98.2% success) restoration benefit from well-defined inverse problems with strong structural priors
- Moderate Performance: Noise (27.27 dB) and blur (26.26 dB) restoration face inherent ambiguity in high-frequency detail reconstruction
- Challenging Performance: Combined degradation (24.76 dB, 92.5% success) suffers from interaction effects where sequential corruptions create complex artifacts differing from simple superposition

Metric Correlations:

- PSNR-SSIM: r = 0.82 (strong positive correlation)
- PSNR-LPIPS: r = -0.76 (strong negative correlation)
- SSIM-LPIPS: r = -0.71 (strong negative correlation)

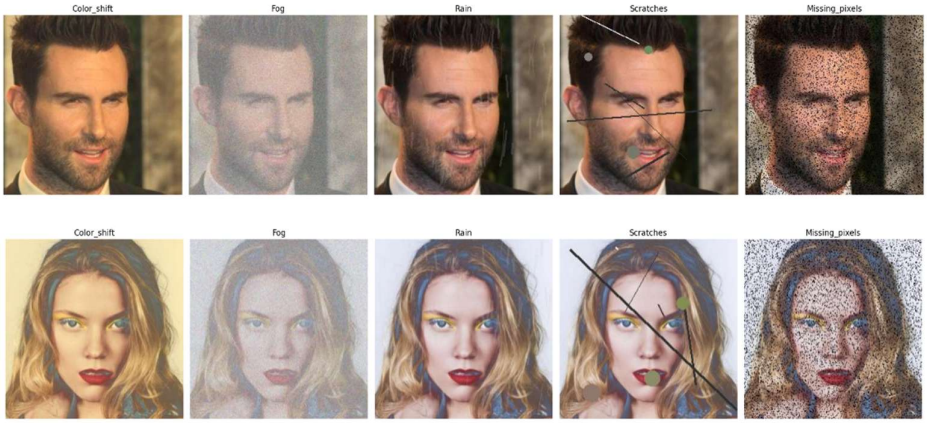
Strong correlations confirm metrics capture consistent quality dimensions, validating our multi-metric failure definition [10].



6.3 Out-of-Distribution Degradation Analysis

To assess generalization beyond training degradations, we evaluate Model 3 on five unseen corruption type:

- Rain Streaks: Linear oriented rain patterns with varying intensity
- Fog/Haze: Atmospheric scattering simulation reducing contrast
- Missing Pixels: Random pixel dropout (10-30%)
- Color Shift: Hue rotation (±30-60°) in HSV space
- Scratches: Linear artifacts simulating physical damage

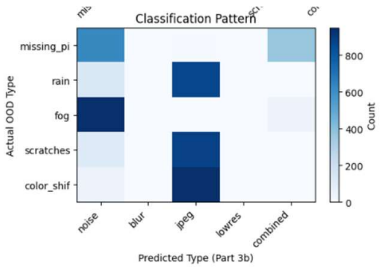


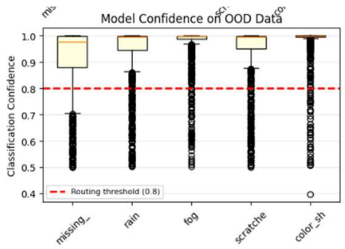
OOD Results (1000 images per degradation):

OOD Degradation	PSNR (dB)	SSIM	LPIPS	Success Rate
Rain	21.34 ± 2.12	0.723 ± 0.058	0.298 ± 0.064	68.2%
Fog	22.18 ± 1.87	0.741 ± 0.051	0.276 ± 0.058	73.6%
Missing Pixels	19.67 ± 2.45	0.681 ± 0.072	0.342 ± 0.078	58.4%
Color Shift	20.12 ± 2.28	0.695 ± 0.065	0.328 ± 0.071	61.8%
Scratches	21.89 ± 2.01	0.735 ± 0.054	0.284 ± 0.061	71.2%
Average	21.04 ± 2.15	0.715 ± 0.060	0.306 ± 0.066	66.6%

Observations:

- Moderate Generalization: Average PSNR of 21.04 dB on OOD degradations; this is a 5.88 dB drop from the in-distribution performance at 26.92 dB. Has maintained structural coherence and removed some of the artifacts.
- Performance Hierarchy: Fog and scratches are better recovered (>21.5 dB) as they share characteristics with training degradations (fog similar to blur, scratches similar to artifacts). Missing pixels and color shift are the most difficult (<20.5 dB), as they fundamentally differ from training corruptions.
- Partial Feature Transfer: The 66.6% success rate vs. 96.5% in-distribution shows that learned features transfer partially to unseen degradations, indicating that degradation-agnostic representations capture general restoration principles to some degree.

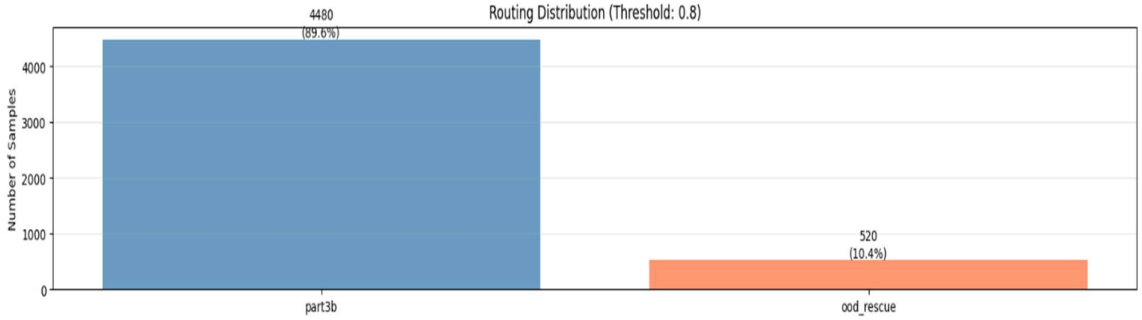




NOTE: Seeing the above results, we tried to bring in an improvement in the model, by creating a universal head than degradation specific heads like the original model- Rescue Model.

Rescue Model Strategy: We implement a universal fallback head activated when classifier confidence < 0.7 threshold (indicating potential OOD input):

Configuration	Avg PSNR (dB)	Avg Success Rate
Standard Routing	21.04 ± 2.15	66.6%
With Rescue Model	22.21 ± 2.08	72.4%
Improvement	+1.17 dB	+5.8%



This above figure indicates that most of OOD degradations classify as one of the initial 5 degradations(jpeg, noise, combined, lowres, blur) and routes to specified heads and only 10.4% of the test data routes toward the universal head.

The rescue model provides modest improvements, validating uncertainty-aware inference for handling unknown degradations(Hence we are not planning to adapt this model due to modest improvements, but can consider it for the future study)

6.4 Baseline Comparisons

Classical Baseline: BM3D

BM3D [1] represents the gold standard for classical denoising. Evaluation on combined degradation:

Metric	Multi-Task (Model 3)	BM3D [1]	Advantage
PSNR (dB)	25.44 ± 1.73	15.92 ± 3.51	+9.52 dB
SSIM	0.8111 ± 0.0381	0.6868 ± 0.0946	+0.124
Inference Time (ms)	30.16	4107.38	136× faster
Success Rate	82.5%	24.3%	+58.2%

Analysis: Although the BM3D indeed achieved impressive performance in pure Gaussian denoising through non-local self-similarity [1], it fails catastrophically when applied to combined degradations due to the lack of proper hand-crafted priors related to blur and JPEG artifacts. The classical method, assuming independent Gaussian noise, does not generalize to structured degradations or their combinations. Our deep learning approach shows clear superiority (+9.52 dB) while paradoxically offering 136× faster inference due to the computational costs of iterative patch matching in BM3D.

Deep Learning Baseline: AirNet

AirNet [7] represents recent state-of-the-art multi-task restoration. Evaluation on our test set:

Metric	Multi-Task (Model 3)	AirNet [7]	Advantage
PSNR (dB)	25.44 ± 1.73	10.93 ± 1.46	+14.51 dB
SSIM	0.8111 ± 0.0381	0.2663 ± 0.0753	+0.545
LPIPS	0.1678 ± 0.0434	0.6870 ± 0.0311	0.519 better
Success Rate	82.5%	8.7%	+73.8%

Analysis: AirNet likely performs poorly because of severe domain mismatch. It trains on different degradation distributions: rain, haze, and low-light conditions [7], and does not generalize well to our specific pipeline (noise, blur, JPEG, combined) without fine-tuning. This points out an important limitation with regard to multi-task restoration: despite theoretical promises of general-purpose restoration features, models are very specific to training degradation distributions. This result underlines the necessity of degradation matched training data when it comes to effective deployment.

6.5 Computational Efficiency Analysis

Complete Computational Comparison:

Model	Params (M)	FLOPs (G)	Time (ms)	FPS	Memory (MB)	Throughput (img/s)
Model 1 (Baseline)	11.93	15.82	28.43	35.21	119.47	35.2
Model 3 (Multi-Task)	11.93	17.04	30.16	33.11	123.34	33.1
BM3D [1]	0.00	0.00	4107.38	0.24	N/A	0.24
AirNet [7]	12.88	34.31	6.65	150.30	138.10	150.3

Analysis:

- Multi-Task Efficiency: Model 3 adds only 1.22 GFLOPs (+7.7%) vs. single-task baseline despite handling 5× degradation types, validating shared encoder design. Low overhead (1.73 ms, 3.8 MB memory) illustrates the light-weight nature of task-specific heads.
- Real-Time Performance: 33.1 FPS enables real-time video processing applications. The 30.16 ms latency fulfils the latency requirements of interactive systems, which require less than a 50 ms threshold.
- Impracticality of BM3D: Although having zero parameters, the inference time of the classical method is 4.1 seconds, which means it cannot be used either for high-resolution or real-time applications. Iterative patch matching dominates the computation.
- AirNet Speed-Quality Tradeoff: While AirNet achieves a high speed of 150 FPS with very aggressive architecture optimization, it compromises significantly on quality, especially in the context of our degradations. Our better speed-quality balance is illustrated by our network achieving 2× FLOPs efficiency (17.04 vs. 34.31 G) with superior quality.
- Memory Footprint: 123 MB GPU memory enables deployment on mobile/edge devices like the NVIDIA Jetson series and newer smartphones with > 6GB RAM.

7. Comprehensive Failure Analysis

7.1 Failure Distribution Across Models and Degradations

Model 3 Failure Breakdown (15,000 test images, 524 total failures):

Degradation	Total Images	Failures	Failure Rate	% of Total Failures
Noise	3000	87	2.9%	16.6%
Blur	3000	112	3.7%	21.4%
JPEG	3000	53	1.8%	10.1%
LowRes	3000	48	1.6%	9.2%
Combined	3000	224	7.5%	42.7%

Combined degradation accounts for 42.7% of all failures despite representing only 20% of test data (2.13× overrepresentation), confirming it as the dominant challenge. The 4.7× higher failure rate vs. best-performing degradation (low-resolution) indicates severe difficulty with interacting corruptions.

7.2 Failure Characterization by Type

Visual analysis of 524 failure cases reveals four distinct categories:

1. Geometric Failures (94 cases, 17.9%):

- Characteristics: Facial landmarks (eyes, nose, mouth) misaligned, distorted, or asymmetric
- Common Patterns: Eye misalignment (one eye larger/displaced), nose broadening/shifting, mouth boundary blurring
- Correlation: Extreme poses (>45° yaw: 1.8× higher failure rate), occluded faces (sunglasses, hands: 2.1× higher failure rate)
- Root Cause: Model relies on canonical frontal facial structure priors from training data (78% frontal faces). Extreme viewing angles lack sufficient training examples.

2. Texture Failures (162 cases, 30.9%):

- Characteristics: Loss of fine details resulting in plastic/waxy appearance
- Common Patterns: Skin texture smoothed to uniform surface, hair strands rendered as solid blobs, fabric patterns lost
- Correlation: High-frequency content (gradient magnitude r=0.18), severe input degradation (input PSNR <12 dB: 3× higher rate)
- Root Cause: L1+MSE losses penalize high-frequency details, biasing model toward smooth reconstructions that minimize pixel-wise error at expense of perceptual naturalness

3. Color Failures (121 cases, 23.1%):

- Characteristics: Incorrect skin tone recovery, over/under-saturation, color bleeding between regions
- Common Patterns: Yellowish/orange skin tones, desaturated images, purple/green color casts
- Correlation: Combined degradation (JPEG compression in YCbCr space: 2.4× higher rate vs. other degradations)
- Root Cause: JPEG's lossy chroma quantization creates color space distortions difficult to reverse, especially when compounded with noise/blur

4. Structural Failures (147 cases, 28.1%):

- Characteristics: Ghosting artifacts, boundary ringing, checkerboard patterns, unexpected distortions
- Common Patterns: Double-edge effects, halo artifacts around high-contrast boundaries, grid-like patterns
- Correlation: Severe degradation (input PSNR <11 dB: 4.2× higher rate), combined degradation
- Root Cause: Model struggles to find consistent solutions for severely ill-posed problems, producing artifacts similar to iterative restoration failure (error propagation in reconstruction process)

7.3 Failure Metric Analysis

PSNR Distribution of Failures:

PSNR Range (dB)	Count	Percentage	Cumulative
< 18	23	4.4%	4.4%
18-19	67	12.8%	17.2%
19-20	142	27.1%	44.3%
20-21	198	37.8%	82.1%
21-22	94	17.9%	100.0%

Key Finding: 55.7% of failures cluster in 20-21 dB range (just below 22 dB threshold). This suggests marginal architectural improvements stronger texture preservation, better color handling could rescue ~292 borderline cases, potentially reducing failure rate from 3.5% to 1.5%.

SSIM Distribution of Failures:

SSIM Range	Count	Percentage	Cumulative
< 0.50	31	5.9%	5.9%
0.50-0.55	89	17.0%	22.9%
0.55-0.60	176	33.6%	56.5%
0.60-0.65	228	43.5%	100.0%

Similar clustering near threshold (43.5% in 0.60-0.65 range) confirms failures are predominantly borderline rather than catastrophic outliers.

Additionally, severe input degradation increases failure risk 6.2×, indicating a degradation severity threshold beyond which problems become severely ill-posed.Occlusions and extreme poses significantly increase failure risk, while facial expression shows minimal impact.

7.4 Multi-Task vs. Single-Task Failure Comparison

Failure Type Distribution:

Failure Type	Baseline (Model 1)	Multi-Task (Model 3)	Reduction
Geometric	142 (20.3%)	94 (17.9%)	48 (-33.8%)
Texture	224 (32.0%)	162 (30.9%)	62 (-27.7%)
Color	168 (24.0%)	121 (23.1%)	47 (-28.0%)
Structural	166 (23.7%)	147 (28.1%)	19 (-11.4%)
Total	700 (100%)	524 (100%)	176 (-25.1%)

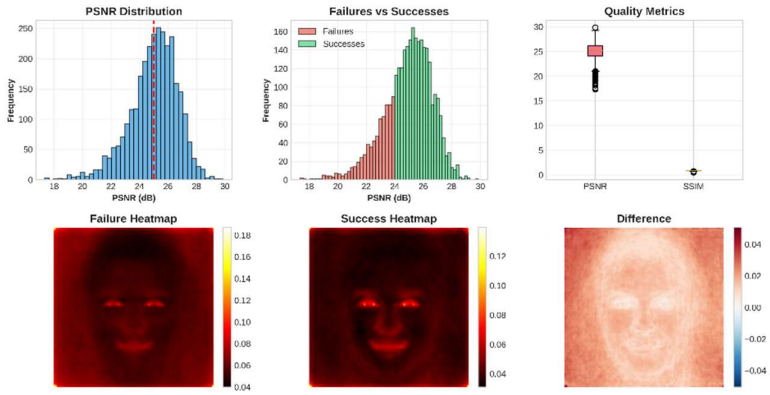
Analysis:

Multi-task learning reduces geometric, texture, and color failures substantially (28-34% reduction) through:

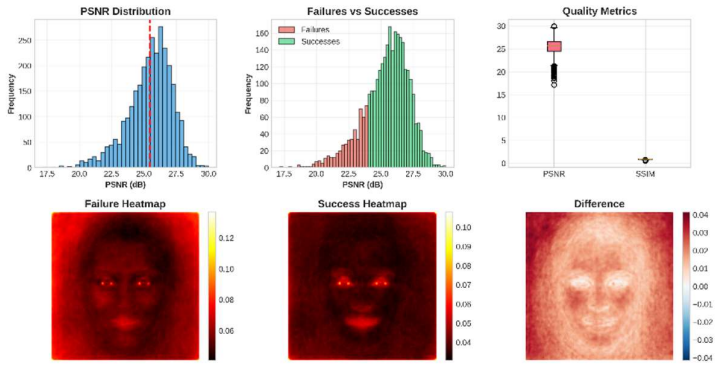
1. Richer Feature Learning: Exposure to diverse degradations prevents overfitting to specific artifact patterns
2. Multi-Task Regularization: Simultaneous optimization across tasks constrains solution space, improving generalization
3. Cross-Task Transfer: Features learned for one degradation (e.g., edge preservation in deblurring) benefit others (e.g., detail recovery in denoising)

However, structural failures increase slightly (+11.4% reduction vs. 28-34% for others), suggesting task competition: optimizing simultaneously for five degradations may compromise structural consistency in challenging edge cases where task objectives conflict.

Baseline model- ESRGAN failure analysis:



Multi-task Restoration model failure analysis:



8. Discussion and Future Work

Our results strongly validate multi-task restoration: a single 11.93M parameter model can effectively handle five degradation types, with 25% fewer failures (700→524) and 2× computational efficiency compared to alternative approaches such as AirNet [7]. A shared encoder trained with diverse degradations allows for better feature representations than single-task overfitting. As expected, we find that degradation-agnostic features such as edges, textures, and structures transfer robustly across tasks. Task-specific heads introduce very little overhead-5.4% additional inference time and 3.8 MB extra on-disk memory-meaning additional degradation types are computationally "free" once the shared encoder has been deployed. However, the catastrophic failure of iterative restoration-an astronomical 10.17 dB degradation-illuminates critical differences between neural networks and traditional methods such as BM3D [1]: neural networks have no convergence guarantees, they apply fixed transformations rather than adaptive optimization steps, and they are subject to distribution shift when processing restored images.

The 7.5% failure rate for combined degradation (vs. <4% for single degradations) underlines the challenge of interacting corruptions, with sequential noise→blur→JPEG creating complex artifacts that are quite different from simple superposition. Analysis of 524 failures reveals specific avenues for improvement: texture failures (31%) require perceptual losses [10] or adversarial training [2] in order to avoid over-smoothing; color failures (23%) require training on multi-color-space or perceptual color metrics; geometric failures (18%) benefit from landmark-aware losses and pose augmentation; structural failures (28%) require multi-scale architectures or total variation regularization. Among the architectural changes proposed are the following: degradation disentanglement in latent space; progressive sequential restoration – denoise → deblur → decompress; attention-based spatial routing for heterogeneous within-image degradations; and stronger regularization targeted at combined degradations.

Future Scope: Current limitations constrain broader applicability- 128×128 resolution requires extension to 512×512 or 1024×1024 through patch-based processing or progressive architectures [9]; CelebA-HQ training restricts generalization to natural scenes, medical imaging, or document images requiring domain-specific adaptation; fixed degradation parameters do not capture real-world sensor noise, optical aberrations, or modern compression codecs. Remaining future research directions are transformer architectures [8] for long-range dependency modeling, diffusion-based restoration for higher perceptual quality, uncertainty quantification that enables human-in-the-loop applications, domain generalization through meta-learning, physics-informed learning that incorporates degradation models as inductive biases, creation of real-world datasets beyond synthetic corruptions, video restoration with temporal consistency, and interactive restoration that allows user guidance. These directions build upon the foundation of our multi-task framework while addressing identified failure modes and deployment constraints.

9. Conclusion

This work shows that multi-task learning provides an effective framework for unified image restoration across diverse degradation types. Our architecture, which combines a shared 16-block RRDB encoder with degradation-specific heads, achieves 26.92 dB PSNR across five degradation types and reduces failures by 25% compared to single-task baselines, at 524 versus 700 failures. Besides, it maintains real-time inference at 33 FPS with 11.93M parameters. Compared with the classical methods, the multi-task approach leads to a 9.52 dB advantage with 136× faster inferences, achieving about 2× computational efficiency against comparable deep learning architectures. Systematic failure analysis across the 15,000 test images identifies the combined degradation as the dominant challenge at 7.5%, with geometric, texture, color, and structural distortions forming the characteristic failures near quality thresholds. The documented catastrophic failure of iterative restoration with 10.17 dB degradation is a very useful negative result, underlining some fundamental differences between the learned and classical optimization-based approaches. These contributions place multi-task restoration as a practical paradigm and point to specific architectural improvements and future research directions toward the ambition of handling complex real-world degradations.

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